

Name	O-Band	E-Band	S-Band	C-band	L-Band
Wavelength range (nm)	1260-1360	1360-1460	1460-1530	1530-1565	1565-1625
C-band system				35 nm	
C+L-band system				95 nm	
Multi-Band System	365 nm				

Fig. 5: Multiband system (Source: <https://www.hhi.fraunhofer.de/en>)

Applications

The applications of multicore fibre and multiband optical networks are extensive, including:

Data centres. MCF and multiband optical networks provide high-density, high-speed connectivity between in-house devices, such as servers, switches, and storage devices

Multicore fibre lasers, amplifiers, and optical sensors. These optical instruments can be designed with high accuracy, high resolution, compactness, stability, reproducibility, and reliability using MCF technology

Biomedical sensing and imaging.

Widely used in medical applications, such as imaging and diagnostics, to provide high-speed and high-resolution data for testing

Experimental physics. MCF is used for testing various phenomena in fields such as quantum mechanics, the Anderson effect, and fluid dynamics experiments

Passive optical networks (PON). MCF helps increase fibre density and addresses cable size limitations and duct congestion issues in PON. It can be used between the optical network terminal (ONT) and optical line terminal (OLT) of an optical

distribution network (ODN)

Singal processing. MCF is used for various signal processing operations, including pulse shaping, modal conversion, and dispersion engineering

Challenges associated

Every new technology comes with both advancements and challenges. Major challenges in MCF and multi-band optical networks include:

Crosstalk between cores. Signals from one core can interfere with neighbouring cores, potentially causing data loss

Need for advanced signal processing. Transmitted data from MCF or multiband networks must be received separately at the receiver's side, which involves complex signal processing to mitigate crosstalk and ensure reliable data transmission

Limited industry acceptance. MCF cannot be directly spliced with single-mode fibres, limiting its industry adoption

India's 1st Independent Survey



For details www.efy.in/elcina-imrb

